Xedni p=67 frozen comparison

Xedni affine-lift optimizer: frozen p = 67 comparison (2026-11-12)

Update 2026-11-12: the frozen p=67 comparison ran once under an owner waiver of the S3 checkpoint (PROTOCOL.md, “Amendment 2026-11-12”; `runs/run-20261112/`): the exact neighborhood (radii 1, 2, 4) gave the same 56-bit discriminant as Babai (radius 0), a bounded null result `[pcc:unverified]`. The text below is the 2026-11-02 state.

The frozen p=67 comparison had **not run** as of 2026-11-02. The mandatory pre-run checkpoint upload is denied by both the installed presigned token (HTTP 403) and the 2026-11-02 managed AWS identity (`PutObject AccessDenied`). The exact blocker and exit condition are recorded in `problems/xedni-affine-cvp-checkpoint-access.md`. Consequently this report makes no empirical claim that the new nonlinear search beats the 2000 implementation.

Claim labels used below are **source**, **derived**, **verified control**, **implemented but unmeasured**, and **blocked**.

What “the nine points” actually means

Source. JKSST Section 3.2 chooses an integer `r` with `2 <= r <= 9`, saying that `r=4..6` is most likely. A general projective cubic has ten coefficients up to the incidence equations, so nine independent point rows are the maximum the construction permits; the algorithm does not require a special fixed set of nine points.

For `r>=4` in general cubic form, the paper first sends four points to a fixed projective frame. Steps 6–7 lift the remaining point coordinates and solve an affine coefficient system. Steps 9–10 vary kernel vectors to reduce the discriminant and apply a rank heuristic. The authors’ implementation already used the Havas–Majewski–Matthews HNF algorithm. Therefore replacing a raw integer solve by “some lattice reduction” is not itself a new improvement over their implementation.

Implemented algorithms

Let `u*` be one feasible coefficient vector, let the rows of `D` be admissible coefficient changes, let `B` be the lifted-point incidence matrix, and let `q=Mp` be the required modulus. The input certificate checks

```
B D_i = 0                 and                 D_i = 0 (mod q)
```

for every direction. The optimizer then:

Protocol amendment

Amendment 2026-11-12: owner waiver of the S3 resume checkpoint, and first run

Appended after the freeze. Every parameter above (fixture, radii, caps, acceptance rules, reproduction command) is unchanged. sha256 of this file before the amendment: `92fef770f86e38fe57e6fbbbce3e48d97412c200c0cc65272aa0049dec7f3620`.

Owner waiver (2026-11-12, given in the crypto session), verbatim: “for this toy-size run only, the S3 resume-checkpoint requirement is waived because no upload credential exists in this container.” The waiver covers the S3 resume-checkpoint requirement for this p=67 run and the sweep in `research/xedni_lift_success_20261112/` only. It does not waive the benchmark record, this frozen protocol, or the certificates. The checkpoint gate was not run and no `CHECKPOINT_SKIP` was used for it.

Run (2026-11-12). Command exactly as in “Reproduction”, from commit `0d69fc5526c02c4071db896d8800be74929f4ff7`, `ca-icx` built in release mode (sha256 `53addfe29282034d2715b33e65edd686b24a2e79e9a20e91dd1c3f71853abb25`), output `runs/run-20261112/`. All four rows completed and satisfied acceptance rules 1 to 5. Discriminant bits (raw / Babai / optimized) and coefficient bits (raw / Babai / optimized):

radius	candidates	disc bits raw / Babai / opt	coeff bits raw / Babai / opt
0	1	1114 / 56 / 56	142 / 15 / 15
1	27	1114 / 56 / 56	142 / 15 / 15
2	125	1114 / 56 / 56	142 / 15 / 15
4	729	1114 / 56 / 56	142 / 15 / 15

Outcome under the protocol’s stop rule: no smaller discriminant at radius r>0 than at radius 0, so this is a bounded null result for the nonlinear neighborhood on this one coset. LLL+Babai alone reduced the raw representative from 1114 to 56 discriminant bits; that is the linear control, not an HMM-HNF comparison. [pcc:unverified]

Run summary


```
        {
          "binary": "/home/user/cryptanalysis/suite/target/release/ca-icx",
          "binary_sha256":
"53addfe29282034d2715b33e65edd686b24a2e79e9a20e91dd1c3f71853abb25",
          "claim_scope": "one deterministic toy prime-field coefficient coset; no
timing or ECDLP-success claim",
          "input":
"/home/user/cryptanalysis/research/xedni_lift_optimizer_20261101/inputs/p67_balanced_snf.json
          "input_sha256":
"65fac6b0b070e9f4205ba0d9be342861b6b12aad95a0c2f15293bc2ff84680db",
          "platform": "Linux-6.18.44-fc-v378-x86_64-with-glibc2.39",
          "python": "3.13.18",
          "rows": [
            {
              "babai_coefficient_bits": 15,
              "babai_discriminant_bits": 56,
              "candidate_cap": 1,
              "candidates_evaluated": 1,
              "congruence_preserved": true,
              "exact_constraints_preserved": true,
              "optimized_coefficient_bits": 15,
              "optimized_discriminant_bits": 56,
              "radius": 0,
              "raw_coefficient_bits": 142,
              "raw_discriminant_bits": 1114,
              "raw_file": "radius-0.json"
            },
            {
              "babai_coefficient_bits": 15,
              "babai_discriminant_bits": 56,
              "candidate_cap": 27,
              "candidates_evaluated": 27,
              "congruence_preserved": true,
              "exact_constraints_preserved": true,
              "optimized_coefficient_bits": 15,
              "optimized_discriminant_bits": 56,
```

```
  "radius": 1,
  "raw_coefficient_bits": 142,
  "raw_discriminant_bits": 1114,
  "raw_file": "radius-1.json"
},
{
  "babai_coefficient_bits": 15,
  "babai_discriminant_bits": 56,
  "candidate_cap": 125,
  "candidates_evaluated": 125,
  "congruence_preserved": true,
  "exact_constraints_preserved": true,
  "optimized_coefficient_bits": 15,
  "optimized_discriminant_bits": 56,
  "radius": 2,
  "raw_coefficient_bits": 142,
  "raw_discriminant_bits": 1114,
  "raw_file": "radius-2.json"
},
{
  "babai_coefficient_bits": 15,
  "babai_discriminant_bits": 56,
  "candidate_cap": 729,
  "candidates_evaluated": 729,
  "congruence_preserved": true,
  "exact_constraints_preserved": true,
  "optimized_coefficient_bits": 15,
  "optimized_discriminant_bits": 56,
```